Fiscal Year:	FY 2008	**Task Last Updated:**	FY 09/29/2008
PI Name:	Zhao, Yongliang Ph.D.		
Project Title:	Functional Role Of The Betaig-H3 Gene In High-Energy Heavy Ions-Induced Carcinogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10032	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	07/01/2003	**End Date:**	06/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 4/2009) NOTE: Received NCE to 06/30/2008 per A. Chu/ARC (6/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shao, Genze (Columbia University) Berenguer, Jessica (Columbia University)		
Grant/Contract No.:	NAG2-1637		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Understanding the cellular and molecular mechanism of radiation carcinogenesis induced by high-energy heavy ions is important for both space radiation protection as well as design of interventional approaches. Our previous data demonstrate that loss of expression of Betaig-h3 gene is causally linked to tumorigenic process in high-LET, alpha particle-irradiated human papillomavirus-immortalized human bronchial epithelial (BEP2D) cells. Furthermore, loss of Betaig-h3 expression is common in many human tumor cell lines and reintroduction of this gene into human lung tumor cells completely abrogates their tumorigenicity, suggesting that Betaig-h3 gene may function as a tumor suppressor in a variety of human tumors. The newly established telomerase (hTERT)-immortalized human bronchial epithelial (hTERT-SAEC) cell model that has normal p53 and p16 functions represents an attractive alternative to further ascertain the mechanism involved in radiation carcinogenesis. In this study, we propose to examine the functional role of Betaig-h3 gene in high-energy heavy ions-induced carcinogenesis using the hTERT-SAEC cells and Betaig-h3 knock-out mice. The proposal has 3 main aims: I) To establish the tumorigenic cell model by irradiating hTERT-SAEC cells using beams of charged heavy ions; II) To demonstrate that loss of Betaig-h3 expression is causally related to the malignant transformation process; and III) To generate mouse embryonic stem cells and knock-out mice with disrupted Betaig-h3 gene. Using the Betaig-h3 null mice model, we will address the following questions: (1) whether mice with disruption of this gene predispose to cancer; (2) Is the incidence of transformation induced by heavy ions higher in Betaig-h3 null embryo fibroblast cells than in wild type cells? (3) Does loss of Betaig-h3 gene in embryo fibroblast cells affect radiation sensitivity and apoptosis induced by high-energy heavy ions? I) To establish the tumorigenic cell model by irradiating hTERT-SAEC cells using beams of charged heavy ions; II) To demonstrate that loss of Betaig-h3 expression is causally related to the malignant transformation process; and III) To generate mouse embryonic stem cells and knock-out mice with disrupted Betaig-h3 gene. The data generated from the proposed studies will provide important insights into the role of Betaig-h3 gene in heavy ion-indued carcinogenesis
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cancer induction by space radiation is a major concern for manned space exploration. Among the various particulate components of ionizing radiation in space, high energy (HZE) heavy ions have been of special concern to radiobiologists and NASA administrators mainly because they contribute about 1% of the particle radiation field in frequency whereas they produce a considerable amount of the whole dose-equivalent because of their large energy deposition and high LET. For the acute effects like cell inactivation are of not so direct interest, late effects including genetic changes and cancer formation are more important for risk estimation. However, for this biological end-point there is no direct human evidence available. The research from this funding project will provide the important information ragarding the malignant transformation and molecular alterations induced by heavy ions.
Task Progress:	High-energy (HZE) heavy ions, when compared to low-LET radiation, are highly effective in inducing gene mutation, chromosomal aberrations and neoplastic transformation. However, the underlying molecular mechanisms are not clearly understood. We have shown previously that the down-regulation of Betaig-h3 expression is causally linked to the tumorigenic phenotype of papillomavirus-immortalized human bronchial epithelial (BEP2D) cells treated with high-LET ?-particle radiation. Using this BEP2D cell culture system, malignantly transformed cell lines have been established by a single dose of 56Fe heavy ion radiation. We found that expression level of the Betaig-h3 gene is markedly decreased in three tumorigenic cell lines compared to parental BEP2D cells. Reconstitution of its expression in tumorigenic cells by transfection of expression vectors significantly suppressed their tumorigenicity. To further define the role of Betaig-h3 in the tumorigenic process, mouse embryonic fibroblasts (MEFs) with Betaig-h3-null and wild type genotypic backgrounds were irradiated by a single dose of 30 cGy or 60 cGy of heavy ions, transformation frequency were then determined in the irradiated cells. The results showed that transformation frequency in Betaig-h3 null MEFs is significantly higher than that in wild type MEFs, and displays a dose-dependent manner. This data further support the hypothesis that Betaig-h3 gene possesses anti-tumor function in heavy ion-induced carcinogenesis. Finally, spontaneous tumor development has been investigated in mice with Betaig-h3-null, heterozygous and wild type genotypic backgrounds. The preliminary data demonstrated that mice with Betaig-h3 deficiency are predisposed to both spontaneous tumors and 7,12-dimethylbenz(a)anthracene (DMBA)-induced skin tumors, providing the strong evidence that Betaig-h3 functions as a tumor suppressor in vivo and further pointing the potential significance of Betaig-h3 loss in process of heavy-ion induced carcinogenesis.
Bibliography Type:	Description: (Last Updated: 09/29/2008)
Abstracts for Journals and Proceedings	Zhang Y, Shao G, Berenguer J, Hei TK, Zhao YL. "Betaig-h3 Gene Functions as a Tumor Suppressor in Heavy Ion-induced Carcinogenesis." NASA Investigators' workshop, League City, Texas, Feb 4-6, 2008. NASA Investigators' workshop, League City, Texas, Feb 4-6, 2008. , Feb-2008
Articles in Peer-reviewed Journals	Zhang Y, Wen GY, Shao GZ, Lin CS, Bhagat G, Balajee AS, Hei TK, Zhao YL. "Disruption of Betaig-h3 gene induces genomic instability and predisposes the mice to spontaneous tumor growth." Cancer Research, (In Review), October 2008. , Oct-2008
Articles in Peer-reviewed Journals	Calaf GM, Echiburú-Chau C, Zhao YL, Hei TK. "BigH3 protein expression as a marker for breast cancer." Int J Mol Med. 2008 May;21(5):561-8. PMID: 18425347 , May-2008
Articles in Peer-reviewed Journals	Shao G, Balajee AS, Hei TK, Zhao Y. "p16INK4a downregulation is involved in immortalization of primary human prostate epithelial cells induced by telomerase." Mol Carcinog. 2008 Oct;47(10):775-83. PMID: 18383581 , Oct-2008
Articles in Peer-reviewed Journals	Shah JN, Shao G, Hei TK, Zhao YL. "Promoter hypermethylation of a putative tumor suppressor gene, Betaig-h3, is related to locoregional invasiveness in human prostate cancer." BMC Cancer , Accepted, October 2008. , Oct-2008